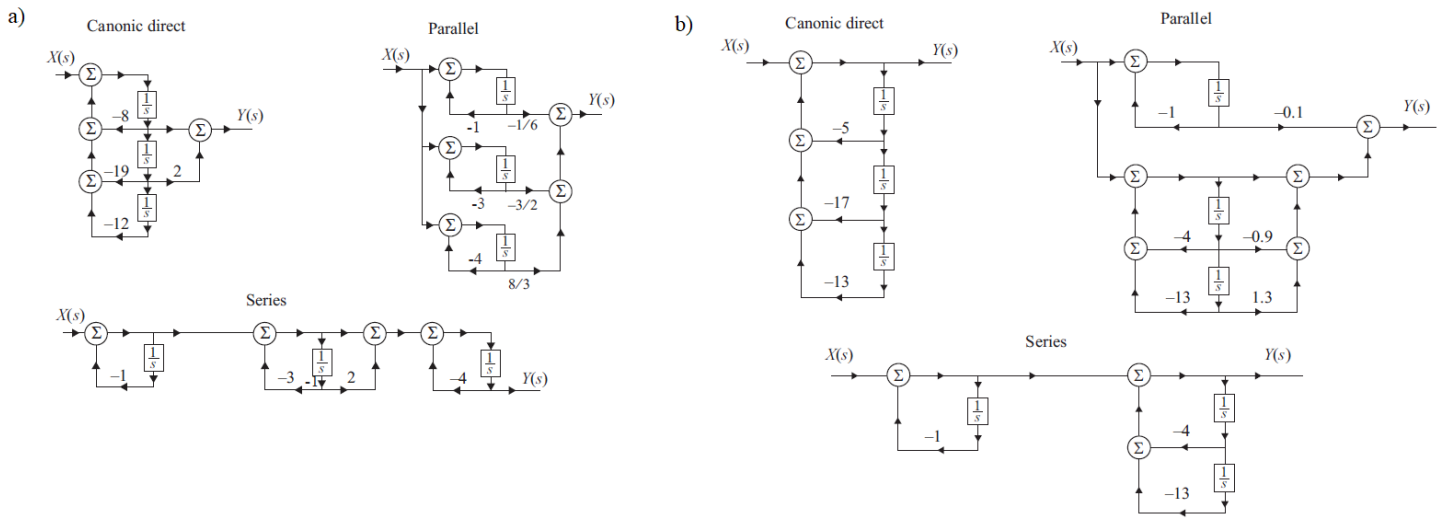


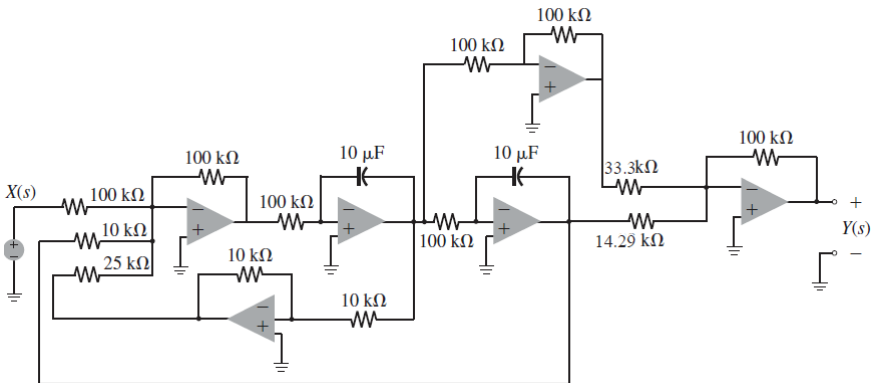
# ENGR 3323: Signals and Systems

## HW 7\_Ch4 Answer Keys

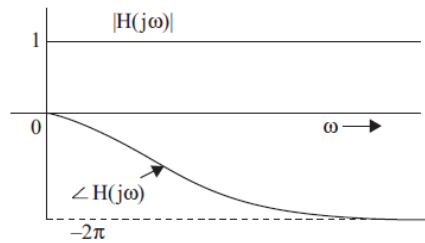
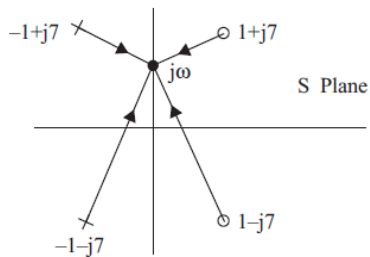
Q1)



Q2)



Q3)

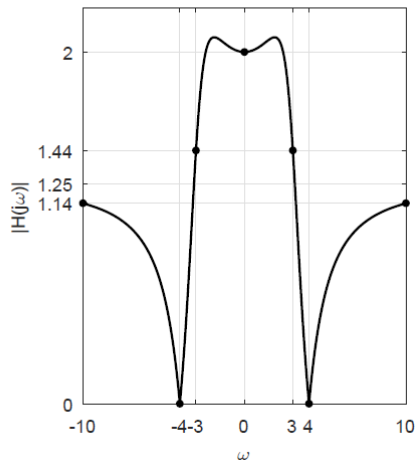


**Q4)**

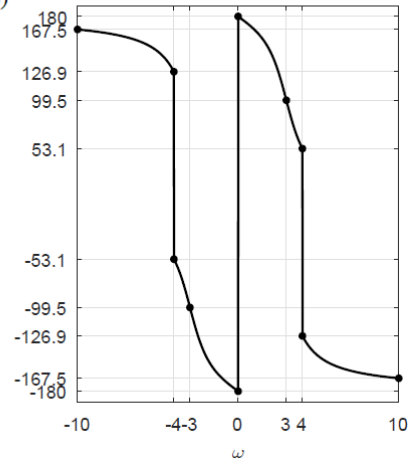
a)  $k = -\frac{5}{4}$ ,  $b_1 = 0$ ,  $b_2 = 16$ ,  $a_1 = 2$ , and  $a_2 = 10$ .

d)  $y(t) = 6 + 1.44 \cos(3t + 159.5^\circ)$

b)



c)



**Q5)** The smaller the  $a$ , the gain will be higher at the vicinity of  $\omega = 10$ .

$$H(s) = \frac{s}{(s + a + j10)(s + a - j10)} = \frac{s}{s^2 + 2as + (100 + a^2)}$$

**Q6)** The feedback system is

$$H(s) = \frac{1}{1 + ae^{-s\tau}}$$